The empirical recurrence for OEIS A223956, proved by transfer matrix

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Abstract

OEIS A223956 counts arrays over $\{0,\ldots,3\}$ whose rows, columns, diagonals or antidiagonals are required to be monotone or unimodal, and carries an empirical recurrence of order 13 contributed by R. H. Hardin. A monotone condition is local, but a unimodal one is not: a sequence is unimodal exactly when it never rises again after it has fallen, so the state must remember which sequences have already fallen. With those bits carried along, $a(n)$ is a walk count on $S = 160$ vertices and the recurrence follows from a finite exact computation.

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1 The sequence and the conjecture

OEIS A223956 is "Number of nX3 0..3 arrays with antidiagonals unimodal and rows and diagonals nondecreasing.". It has offset 1 and begins

$$20, 295, 3471, 41006, 485714, 5777663, 68892600, 822141033, \ldots$$

The entry carries the following comment:

> Empirical: a(n) = 20*a(n-1) -105*a(n-2) +51*a(n-3) +589*a(n-4) +1340*a(n-5) -8833*a(n-6) -2855*a(n-7) +30423*a(n-8) +57774*a(n-9) -140760*a(n-10) -176184*a(n-11) +106272*a(n-12) +544320*a(n-13)

As of the "Last modified" line on the live entry (Oct 03 2025 23:01:34, revision 8) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \;=\; 20\,a(n-1) - 105\,a(n-2) + 51\,a(n-3) + 589\,a(n-4) + 1340\,a(n-5) - 8833\,a(n-6) - 2855\,a(n-7) + 30423\,a(n \tag{1}$$

for all $n > 13$.

2 The conditions

The arrays have 3 columns and $L = n$ rows. Written in (row, column) coordinates, the entry asks:

- along the direction $(1, -1)$ — one step of $(1, -1)$ in (row, column) coordinates — every sequence is unimodal;
- along the direction $(0, 1)$ — one step of $(0, 1)$ in (row, column) coordinates — every sequence is nondecreasing;

- along the direction $(1,1)$ — one step of $(1,1)$ in (row, column) coordinates — every sequence is nondecreasing;

Here “unimodal” means nondecreasing and then nonincreasing, that is: no rise after a fall.

A unimodal condition across the rows is not local: whether a rise is allowed depends on whether the sequence has already fallen. The state therefore carries, for each of the 3 sequences currently running in that direction, one bit saying whether it has fallen. Those bits travel with their sequences — the bit at column u moves to column $u + d_2$ when the next row is laid down, and a sequence beginning at the far end of the new row starts with its bit clear.

Lemma 1. *Let $\mathcal{V}$ be the set of pairs (a row satisfying every within-row condition, a bit vector), $|\mathcal{V}| = S = 160$; let $(r, f) \to (s, f')$ be an edge when placing s after r breaks none of the across-row conditions and f' is f updated by that step; let $\mathbf{v}$ mark the vertices whose bit vector is the one their row alone produces. Then*

$$a(n) \;=\; \mathbf{v}^{\top} N^{\,L-1} \mathbf{1}, \qquad L = n.$$

Proof. A walk of length $L-1$ is a sequence of L rows, that is, an array, and its bit component is determined by them. Every within-row condition is imposed on each vertex; every across-row condition compares two consecutive rows and is imposed on each edge; and the unimodal and lexicographic bits are, by construction, the record of what the rows laid down so far have already decided, so a violation removes the corresponding edge. Hence the walks are exactly the admissible arrays. Nothing constrains the last row beyond what has already been imposed, so the walk may end anywhere. The state carries a row, not a boundary, so L rows make a walk of length $L-1$. □

3 The criterion, and the computation

Let $q(t) = t^{13} - \sum_i c_i\, t^{13-i}$ be the characteristic polynomial of (1) and $G = N$.

Lemma 2. *With n_0 the least index for which $L \geq 1$, $o = 1n_0 + 0 - 1$ and $h_j = G^j N^o \mathbf{1}$, we have for $j \geq 13$*

$$a(n_0 + j) - \sum_i c_i\, a(n_0 + j - i) = \mathbf{v}^{\top} G^{\,j-13} w, \qquad w = h_{13} - \sum_i c_i h_{13-i}.$$

Hence (1) holds from that point on if and only if $\mathbf{v}^{\top} G^m w = 0$ for all $m \geq 0$, and $m < S$ suffices.

Proof. Substitute Lemma 1 and factor. By Cayley–Hamilton G^S is an integer combination of $I, G, \ldots, G^{S-1}$, so every later value repeats that combination of earlier ones. □

Theorem 3. *The recurrence (1) holds for every $n > 13$, which is the statement of Section 1.*

Proof. The digraph was built directly from the conditions listed above. The vector w was formed by 13 applications of G in exact integer arithmetic and $\mathbf{v}^{\top} G^m w$ evaluated for $m = 0, 1, \ldots,$ again exactly. Every one of those integers vanishes from the index corresponding to $n = 13+1$ onwards, and the run continued until $S + 1$ consecutive zeros had been seen. □

4 Verification

Three checks, each independent of the algebra above.

First, the walk counts of Lemma 1 were compared against the entry’s DATA at every one of the 16 published indices, with no shift fitted. The values agree exactly. This is what fixes the reading of the wording: “rows and columns in nondecreasing order” means the rows and columns are sorted as vectors, not that each is nondecreasing along itself, and the two readings already differ at the second published term — 86 against 50 for 2×2 arrays over $\{0, 1, 2, 3\}$. The DATA decides it.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the annihilation test of Lemma 2 was carried to the full Cayley–Hamilton bound in exact integer arithmetic rather than to a sampled prefix.

References

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